

Supplementary Materials for

Comment on “China’s solar expansion policy reduces bird diversity”

Chenchen Ding. Correspondence to: dingchenchen@pku.edu.cn

This PDF file includes: Materials and Methods; Figs. S1 and S2; Tables S1 and S2.

Materials and Methods

Data and exact replication. All analyses start from the replication package released by Zhang *et al.* (Stata code and county-month data; github.com/jianke22/China-s-solar-expansion-policy-reduces-bird-diversity). We rebuilt the estimation panel in R 4.5.1 (fixest, data.table, haven), following the authors’ merge order, variable construction, 1%/99% winsorization, and singleton-dropping conventions. The preferred specification regresses the county-month Shannon index on the policy-stringency index (PSI) and nine controls with county and year-month fixed effects and county-clustered standard errors. Our estimates match Table 1 of the original article to the fourth decimal place (column 3: $\beta = -0.01253$, $SE = 0.00366$, $n = 46,371$), and the headline -2.10% equals $\beta \times \text{s.d. (PSI)}/\text{mean(H)} = -0.0125 \times 4.116/2.446$.

Effort variables and adjusted specifications. The replication data include, for every county-month, the number of birdwatchers (BN) and total birdwatching hours (BT). The authors’ only effort control, **Duration** = $\ln(\text{BT}/\text{BN} + 1)$, is a per-birder ratio: within the estimation sample it correlates only weakly with total effort, whereas summed BN grew 171-fold over 2014–2023. Adjusted specifications add $\ln(\text{BN})$, $\ln(\text{BT} + 1)$, or both to the authors’ model without any other change. Effects are reported as percentages of the outcome mean per 1 s.d. of PSI throughout.

Policy-source restriction and negative-control tests. The policy file covers 1,641 of the 2,344 sample counties; the authors set PSI to zero for the remaining 703 counties (43.9% of observations), although the source itself contains explicit zeros. The restricted sample retains policy-documented counties only. Negative-control regressions replace the outcome with $\ln(\text{BN})$, realized photovoltaic area (converted from installed capacity as in the authors’ code), or $\ln(\text{nightlight} + 1)$; the pre-trend test adds next-year PSI alongside current PSI; a further variant adds county-specific linear trends to the authors’ specification. Results are collected in Table S2.

Data-quality rules. Because Shannon H, Gini–Simpson D, and richness S computed from one assemblage must satisfy $H \leq \ln S$, $D \leq 1 - e^{-H}$ (equivalent to $^2D \leq ^1D$), $D \leq 1 - 1/S$, and $H > 0$

whenever $S \geq 2$, violations identify impossible entries without ecological judgment. Applied to all 46,283 pre-winsorization county-months, the rules flag a single hard violation ($S = 4$, $H = 8 \times 10^{-7}$; Xinhui County, March 2015), which coincides with the 2,147,483,647-egret record verified by the China Ornithological Society, plus 18 county-months carrying the same single-huge-count signature ($S \geq 20$, $H < 0.2$). “Corrupted removed” specifications exclude these 19 county-months.

Recording-protocol simulations. To separate sampling accumulation from genuine community differences, we simulated observation of a fixed lognormal community (60 species, true Shannon $H = 3.19$) under three recording protocols. With counts recorded faithfully, observed H rises with effort and converges to the true value from below. With checklist-type recording (all counts set to one), observed H equals the log of the number of species detected, rises with effort without bound, and exceeds the true value at high effort (4.04 at the highest effort simulated). Adding a single genuine large flock (20,000 individuals) depresses H to 0.02–0.86. A fixed community therefore suffices to generate the observed effort–diversity gradient, and the three protocols yield mutually incompatible “diversity” values from identical communities.

Species-level verification. Through a separate project we hold record-level data from the same platform: the platform’s raw provincial delivery (2024 increment: 32 provinces, 8.1 million species-by-checklist rows, 282,197 checklists) and an event table built from an earlier multi-year delivery (2000–2025, 11.7 million rows). In the raw delivery the count column has zero missingness, and checklists in which a constant count equals the species count (the signature of a list-length leak) comprise at most 0.35% in any province, confirming genuine per-species abundance. Every count equals one in 43.7% of raw-delivery checklists and in 46.9% of 2014–2023 checklists in the event table, consistent with the 40.5% reported platform-wide by the Ornithological Society. Platform deliveries are ceiling-cleaned (maximum count 10^5); the three astronomical records verified by the Society appear in none of them, yet one demonstrably enters the published outcome variable. County-month Shannon recomputed from the event table correlates with the published outcome at $r = 0.71$ across 17,404 overlapping county-months, indicating strong sensitivity of the aggregated index to checklist-set choice.

Specification curve. We estimated every combination of sample (all counties; policy-documented), cleaning (as published; corrupted county-months removed; ≥ 2 birders), outcome metric (Shannon; Simpson; log richness), and effort adjustment (authors’ *Duration* only; explicit total-effort terms), yielding 36 specifications with identical fixed effects and clustering (Table S1; fig. S1). Poisson fixed-effects models for richness with a $\ln(\text{BN})$ offset, following the catch-per-unit-effort convention, appear in Table S2.

Power and minimum detectable effects. For the preferred corrected specification (corrupted records removed, policy-documented counties, explicit effort terms), the minimum detectable effect

at 80% power and $\alpha = 0.05$ is $2.80 \times \text{SE}$, i.e., 1.57% (Shannon), 1.20% (Simpson), and 4.04% (log richness) per 1 s.d. of PSI. The power to detect the published -2.10% , were it real, is $\Phi(0.0125/\text{SE} - 1.96) = 96\%$.

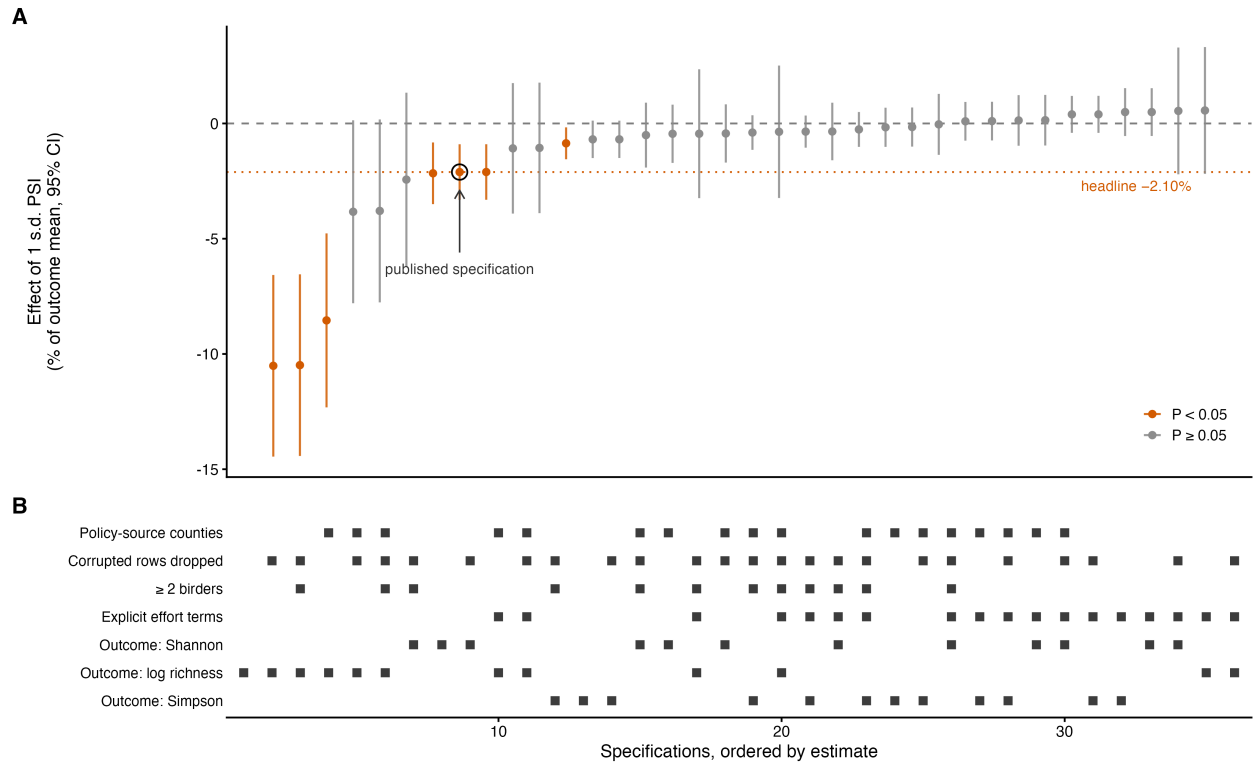


Fig. S1. Specification curve across 36 defensible analysis paths. Estimates of the PSI effect (% of outcome mean per 1 s.d., 95% CIs) for the full cross of sample, cleaning, outcome metric, and effort adjustment, ordered by estimate; the lower panel indicates each path's decisions. All seven significant negative estimates (orange) arise without effort terms; all 18 effort-adjusted paths are null. The open circle marks the published specification.

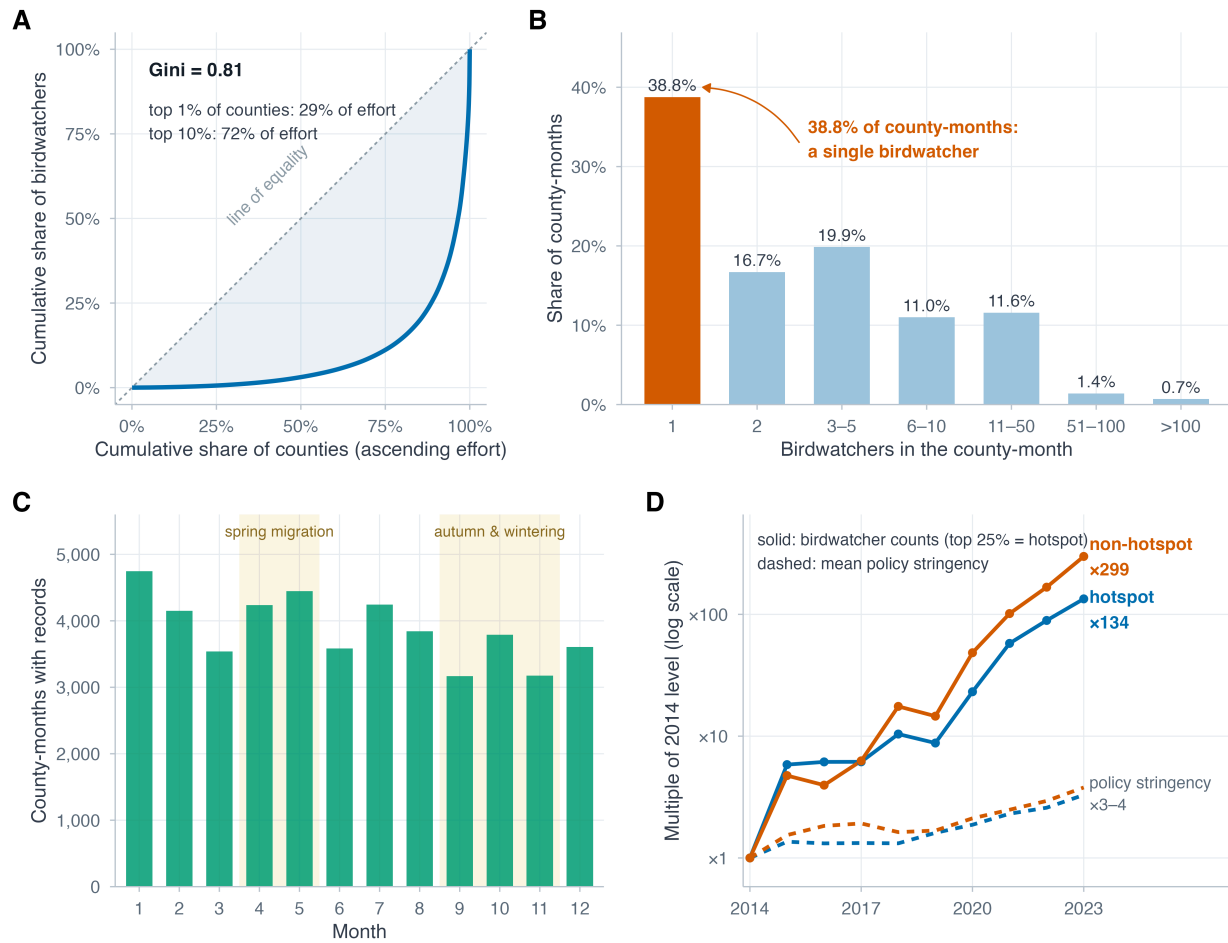


Fig. S2. Structure of the birdwatching data underlying the outcome variable. (A) Observation effort is highly concentrated among counties (Gini = 0.81; 1% of counties contribute 29% of effort). (B) 38.8% of county-months derive from a single birdwatcher. (C) Reporting is strongly seasonal, peaking in wintering and migration periods. (D) Effort growth (solid lines) outpaces policy growth (dashed) by two orders of magnitude, and non-hotspot counties ($\times 299$) outgrow hotspot counties ($\times 134$); the phase mismatch between these diffusion processes underlies the within-county association between policy and effort.

Table S1. Specification curve: all 36 combinations of sample, cleaning, outcome metric, and effort adjustment. All models include county and year-month fixed effects with county-clustered standard errors. Effects are % of the outcome mean per 1 s.d. of PSI; rows ordered by estimate. Rows in bold denote statistically significant negative estimates (all arise without effort terms).

#	Sample	Cleaning	Metric	Effort	Effect (%/s.d.)	95% CI	P	n
1	All counties	As published	log richness	Duration only	-10.51	[-14.45, -6.57]	1.8e-07	45,606
2	All counties	Corrupted removed	log richness	Duration only	-10.48	[-14.42, -6.55]	2e-07	45,588

#	Sample	Cleaning	Metric	Effort	Effect (%/s.d.)	95% CI	<i>P</i>	<i>n</i>
3	All counties	≥2 birders	log richness	Duration only	-8.54	[-12.32, -4.77]	9.7e-06	27,571
4	Policy-documented	As published	log richness	Duration only	-3.83	[-7.80, +0.14]	0.059	25,245
5	Policy-documented	Corrupted removed	log richness	Duration only	-3.79	[-7.76, +0.17]	0.061	25,234
6	Policy-documented	≥2 birders	log richness	Duration only	-2.43	[-6.20, +1.33]	0.21	14,032
7	All counties	≥2 birders	Shannon	Duration only	-2.16	[-3.50, -0.82]	0.0016	28,052
8	All counties	As published	Shannon	Duration only	-2.11	[-3.31, -0.90]	0.00062	46,371
9	All counties	Corrupted removed	Shannon	Duration only	-2.11	[-3.31, -0.90]	0.00062	46,353
10	Policy-documented	As published	log richness	Explicit effort	-1.08	[-3.91, +1.75]	0.45	25,245
11	Policy-documented	Corrupted removed	log richness	Explicit effort	-1.06	[-3.89, +1.77]	0.46	25,234
12	All counties	≥2 birders	Simpson	Duration only	-0.86	[-1.55, -0.17]	0.015	28,052
13	All counties	As published	Simpson	Duration only	-0.69	[-1.50, +0.12]	0.096	46,371
14	All counties	Corrupted removed	Simpson	Duration only	-0.69	[-1.50, +0.12]	0.095	46,353
15	Policy-documented	≥2 birders	Shannon	Duration only	-0.51	[-1.91, +0.90]	0.48	14,510
16	Policy-documented	As published	Shannon	Duration only	-0.45	[-1.71, +0.82]	0.49	26,007
17	All counties	≥2 birders	log richness	Explicit effort	-0.45	[-3.24, +2.35]	0.75	27,571
18	Policy-documented	Corrupted removed	Shannon	Duration only	-0.43	[-1.70, +0.83]	0.5	25,996
19	Policy-documented	≥2 birders	Simpson	Duration only	-0.39	[-1.14, +0.36]	0.3	14,510
20	Policy-documented	≥2 birders	log richness	Explicit effort	-0.36	[-3.23, +2.51]	0.81	14,032
21	All counties	≥2 birders	Simpson	Explicit effort	-0.35	[-1.05, +0.34]	0.32	28,052
22	All counties	≥2 birders	Shannon	Explicit effort	-0.35	[-1.60, +0.90]	0.58	28,052

#	Sample	Cleaning	Metric	Effort	Effect (%/s.d.)	95% CI	<i>P</i>	<i>n</i>
23	Policy- documented	≥2 birders	Simpson	Explicit effort	-0.26	[-1.01, +0.50]	0.5	14,510
24	Policy- documented	As published	Simpson	Duration only	-0.17	[-1.02, +0.68]	0.7	26,007
25	Policy- documented	Corrupted removed	Simpson	Duration only	-0.16	[-1.00, +0.69]	0.72	25,996
26	Policy- documented	≥2 birders	Shannon	Explicit effort	-0.04	[-1.36, +1.28]	0.95	14,510
27	Policy- documented	As published	Simpson	Explicit effort	+0.09	[-0.75, +0.93]	0.83	26,007
28	Policy- documented	Corrupted removed	Simpson	Explicit effort	+0.10	[-0.74, +0.94]	0.81	25,996
29	Policy- documented	As published	Shannon	Explicit effort	+0.13	[-0.97, +1.23]	0.81	26,007
30	Policy- documented	Corrupted removed	Shannon	Explicit effort	+0.14	[-0.96, +1.24]	0.8	25,996
31	All counties	Corrupted removed	Simpson	Explicit effort	+0.39	[-0.41, +1.19]	0.33	46,353
32	All counties	As published	Simpson	Explicit effort	+0.39	[-0.41, +1.20]	0.34	46,371
33	All counties	As published	Shannon	Explicit effort	+0.49	[-0.54, +1.53]	0.35	46,371
34	All counties	Corrupted removed	Shannon	Explicit effort	+0.49	[-0.54, +1.53]	0.35	46,353
35	All counties	As published	log richness	Explicit effort	+0.54	[-2.21, +3.29]	0.7	45,606
36	All counties	Corrupted removed	log richness	Explicit effort	+0.56	[-2.19, +3.31]	0.69	45,588

Robustness variants of the preferred specification (corrupted removed, policy-documented, explicit effort):

Variant	Metric	Effect (%/s.d.)	95% CI	<i>P</i>	<i>n</i>
Province-clustered SEs	Shannon	+0.14	[-1.40, +1.68]	0.86	25,996
Province-clustered SEs	log richness	-1.06	[-5.96, +3.83]	0.67	25,234
Period 2014–2019	Shannon	-3.40	[-7.39, +0.59]	0.095	11,406
Period 2020–2023	Shannon	+0.03	[-1.12, +1.18]	0.96	34,496
Period 2020–2023	log richness	+0.36	[-2.52, +3.23]	0.81	33,904

Table S2. Diagnostic and negative-control regressions. All models include county and year-month fixed effects; standard errors clustered by county.

Regression (county and year-month FE, county-clustered)	β (PSI)	<i>P</i>	<i>n</i>	Note
ln(birdwatchers) on PSI, full sample	-0.0410	2.1e-11	46,371	
ln(birdwatchers) on PSI, policy-documented counties	-0.0078	0.166	26,007	81% attenuation
ln(birdwatchers) on next-year PSI, conditional on current PSI	-0.0447	1.51e-06	32,778	pre-trend
Realized PV area on PSI	-0.0175	0.268	46,371	no exposure link
ln(nightlight) on PSI	-0.0017	0.237	46,371	clean negative control
Shannon on PSI, county-specific linear trends added	-0.0079	0.102	46,371	headline non-significant
Shannon on PSI + ln(birdwatchers)	+0.0032	0.31	46,371	effect null under effort control
Richness (Poisson FE): Poisson FE, no effort (authors' logic)	-0.0233	6.29e-08	45,606	-9.2%/s.d.
Richness (Poisson FE): + offset log(birdwatchers) [CPUE]	+0.0151	0.00568	45,606	+6.4%/s.d.
Richness (Poisson FE): offset + policy-documented counties	+0.0055	0.318	25,245	+2.3%/s.d.
Richness (Poisson FE): offset + ln(hours) + policy-documented	+0.0029	0.553	25,245	+1.2%/s.d.